Supplementary Material

A – Model Performance

Table A1 – Deep learning model performance tested on various sizes of Training data (10 – 100%). Metrics are denoted as estimated mean score ± SD over ten independent training runs for each training data size scenario. Reported metrics are F1-score (micro and macro), categorical accuracy, and area under the curve (AUC).

Training data size	F1-score (micro)	F1-score (macro)	Cat. Accuracy	AUC
100% (final model)	0.92 ± 0.01	0.84 ± 0.01	0.92 ± 0.01	0.97 ± 0.00
90%	0.91 ± 0.01	0.82 ± 0.01	0.91 ± 0.01	0.97 ± 0.00
80%	0.91 ± 0.01	0.82 ± 0.01	0.91 ± 0.01	0.97 ± 0.00
70%	0.90 ± 0.02	0.81 ± 0.02	0.90 ± 0.02	0.97 ± 0.00
60%	0.89 ± 0.01	0.79 ± 0.02	0.89 ± 0.01	0.97 ± 0.00
50%	0.89 ± 0.02	0.79 ± 0.02	0.89 ± 0.02	0.97 ± 0.00
40%	0.88 ± 0.01	0.77 ± 0.02	0.88 ± 0.01	0.96 ± 0.01
30%	0.87 ± 0.02	0.75 ± 0.03	0.87 ± 0.02	0.96 ± 0.01
20%	0.86 ± 0.02	0.73 ± 0.02	0.86 ± 0.02	0.96 ± 0.00
10%	0.86 ± 0.02	0.74 ± 0.04	0.86 ± 0.02	0.96 ± 0.01

Table A2 – Deep learning model performance tested under varying levels of class balance in the training data. Performance metrics are shown as estimated mean ± SD across ten independent training runs for each scenario. Reported metrics include the F1-score (micro and macro), categorical accuracy, and area under the curve (AUC). All models were trained on datasets of equal total size (n = 474) but with different class distributions: “Balanced” (equal proportions of all classes), “Mild” (brooding = 20%, incubation = 40%, random = 40%), “Moderate” (brooding = 15%, incubation = 30%, random = 55%), and “Natural” (brooding = 12%, incubation = 35%, random = 53%; reflecting more natural class imbalance in the dataset).

Class balance scenario	F1-score (micro)	F1-score (macro)	Cat. Accuracy	AUC
Balanced	0.75 ± 0.02	0.70 ± 0.01	0.75 ± 0.02	0.92 ± 0.01
Mild	0.83 ± 0.01	0.75 ± 0.01	0.83 ± 0.01	0.95 ± 0.00
Moderate	0.84 ± 0.01	0.77 ± 0.01	0.84 ± 0.01	0.96 ± 0.01
Natural	0.85 ± 0.01	0.74 ± 0.01	0.85 ± 0.01	0.95 ± 0.01

Table A3 – Deep learning model performance variable combinations. Performance metrics are shown as estimated mean ± SD across ten independent training runs for each scenario. Reported metrics include the F1-score (micro and macro), categorical accuracy, and area under the curve (AUC).

Variable set	F1-score (micro)	F1-score (macro)	Cat. Accuracy	AUC
Light + Tmin + Tmax	0.87 ± 0.01	0.76 ± 0.01	0.87 ± 0.01	0.95 ± 0.00
Raw Light + Tmin + Tmax	0.81 ± 0.01	0.69 ± 0.02	0.81 ± 0.01	0.93 ± 0.00
Light	0.87 ± 0.01	0.78 ± 0.02	0.87 ± 0.01	0.97 ± 0.00
Raw Light	0.87 ± 0.01	0.76 ± 0.01	0.87 ± 0.01	0.95 ± 0.00
Light + Tmin	0.87 ± 0.01	0.79 ± 0.01	0.87 ± 0.01	0.95 ± 0.01
Light + Tmax	0.86 ± 0.01	0.74 ± 0.01	0.86 ± 0.01	0.96 ± 0.01
Raw Light + Tmin	0.83 ± 0.01	0.71 ± 0.01	0.83 ± 0.01	0.95 ± 0.00
Raw Light + Tmax	0.82 ± 0.02	0.68 ± 0.02	0.82 ± 0.02	0.93 ± 0.01